

CONTACT

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 [Muhammad Farhan](#)
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SUMMARY

Data Engineer specializing in building scalable, real-time data pipelines and cloud-native architectures. Expert in Azure, Microsoft Fabric, Databricks, and dbt, with a proven track record of developing metadata-driven frameworks and containerized ELT workflows. Skilled in transforming complex datasets into actionable insights through Medallion architectures and automated orchestration.

TECHNICAL SKILLS

- **Cloud Platforms:** Azure (Databricks, Fabric, Event Hubs, ADF, ADLS Gen2, Functions), AWS (S3, Glue, Athena, Lambda, EMR, EC2).
- **Data Engineering:** Apache Airflow, dbt (Data Build Tool), Apache Spark (PySpark), Medallion Architecture, Delta Lake, Kafka, Microsoft Fabric.
- **Databases & Warehousing:** Snowflake, Databricks Lakehouse, PostgreSQL, Azure SQL, Synapse Analytics.
- **Languages & Tools:** Python (Automation & ETL), SQL, Scala, Docker, uv (Package Management), Git, CI/CD (GitHub Actions).
- **AI & Vector Engineering:** LangChain, RAG , FAISS, Sentence-Transformers, Semantic Search.

WORK EXPERIENCE

Uber Real-Time Data Engineering (Azure & Databricks)

- **The Challenge:** Engineering a high-throughput, low-latency streaming architecture to handle concurrent ride-sharing events while ensuring data integrity across a Medallion architecture.
- **Data & Scale:** Real-time GPS and transaction streams ingested via FastAPI and Azure Event Hubs, processed into Delta Tables.
- **Implementation:**
 - Designed a streaming ETL pipeline in Azure Databricks utilizing Spark Structured Streaming.
 - Implemented Medallion Architecture (Bronze, Silver, Gold) to provide clean, aggregated datasets.
 - Utilized Azure Schema Registry to manage schema evolution across the event-driven ecosystem.

Advanced Azure Data Engineering (Synapse Analytics Integration)

- **The Challenge:** Integrating diverse enterprise data sources into a unified analytics platform while optimizing cost and performance for massive-scale querying.
- **Data & Scale:** Large-scale enterprise datasets moved through Synapse Pipelines and stored in Dedicated SQL Pools for high-performance BI.
- **Implementation:**
 - Architected an end-to-end integration using Azure Synapse Analytics, leveraging Serverless SQL pools for data exploration.
 - Developed complex T-SQL transformation scripts to build dimension and fact tables optimized for analytical workloads.
 - Integrated with Azure Key Vault to manage secure connection strings and credentials across the pipeline.

ShoppingMart Data Warehouse (Microsoft Fabric)

- **The Challenge:** Migrating traditional retail ETL processes to a modern SaaS data lakehouse environment to simplify management and scale-out performance.
- **Data & Scale:** Multi-source retail transaction data ingested into OneLake using Fabric Notebooks and Data Factory.
- **Implementation:**
 - Built a unified data repository in **Microsoft Fabric Lakehouse**, utilizing **Delta Tables** for ACID compliance.
 - Orchestrated data flows using **Fabric Data Pipelines**, automating movement from ingestion to curated Gold layers.
 - Utilized **SQL Analytics Endpoints** within Fabric to enable direct Power BI connectivity for real-time sales reporting.

E-Commerce Data Warehouse & Incremental Analytics Platform

- **The Challenge:** Building a production-grade incremental ETL platform that efficiently processes large e-commerce datasets while delivering analytics-ready data and automated reporting.
- **Data & Scale:** Multi-source Brazilian e-commerce datasets including customers, orders, products, sellers, payments, reviews, and order items processed through a layered warehouse architecture.
- **Implementation:**
 - Developed a containerized data platform using PostgreSQL, Apache Airflow, Docker Compose, pgAdmin, and Apache Superset.
 - Engineered a watermark-based incremental loading framework to process only newly inserted and modified records.
 - Designed a structured warehouse architecture (RAW → STAGING → MARTS) to improve data quality and maintainability.
 - Automated end-to-end ETL workflows using Apache Airflow for extraction, transformation, validation, and loading.
 - Created interactive dashboards in Apache Superset to analyze sales performance, customer behavior, revenue trends, and business KPIs.

Telecommunications Data Lakehouse & Analytics Platform

- **The Challenge:** Designing a scalable telecom lakehouse capable of processing customer activity, call detail records (CDRs), recharge transactions, and mobile data usage while supporting efficient incremental data ingestion and analytics.
- **Data & Scale:** Telecom datasets consisting of customers, calls, recharges, and data usage records ingested into Amazon S3 and transformed through a Medallion Architecture (Bronze, Silver, Gold).
- **Implementation:**
 - Architected an AWS-native Lakehouse using Amazon S3, AWS Glue, PostgreSQL, and Amazon Athena.
 - Developed watermark-based incremental ETL pipelines to process only new and updated records.
 - Implemented Bronze, Silver, and Gold layers for data cleansing, standardization, validation, and business transformations.
 - Built dimensional models and analytical datasets to support customer behavior, revenue, and usage reporting.
 - Leveraged AWS Glue Data Catalog and Athena to enable serverless querying and analytics.

EDUCATION

Superior University, Lahore, Pakistan

- Bachelor Of Computer Science

CERTIFICATES

PySpark – Apache Spark Programming
Udemy

Azure Databricks & Spark For Data Engineer
Udemy